MODSIM-Curvefit

Model Parameter Estimator
for
MODSIM

USER MANUAL

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Models that are used in MODSIM usually include parameters that require calibration against experimental data. This can be done by finding the least-squares best estimates of the parameters that will match some predictions of the models against experimental observations obtained from a laboratory, pilot plant or full scale continuous test. Parameters can be estimated from experiments of this type using the Curvefit software that is available with Modsim.

The strategy that is employed is to embed the entire Modsim simulator inside a modern optimization search engine and have it systematically adjust the chosen parameters to minimize the sum of squared differences between experimental data that is measured on the plant and the simulated counterpart. This is a large calculation but is feasible with Modsim because of its efficient programming methods.

This program requires a two-stage procedure that is easy and convenient to implement.

STAGE 1

The first stage in the estimation process consists in setting up a regular MODSIM job that simulates the combination of units that was used to generate the data on the plant or in the laboratory. The units must be connected in the same configuration as in the physical plant. The feed stream is specified in MODSIM

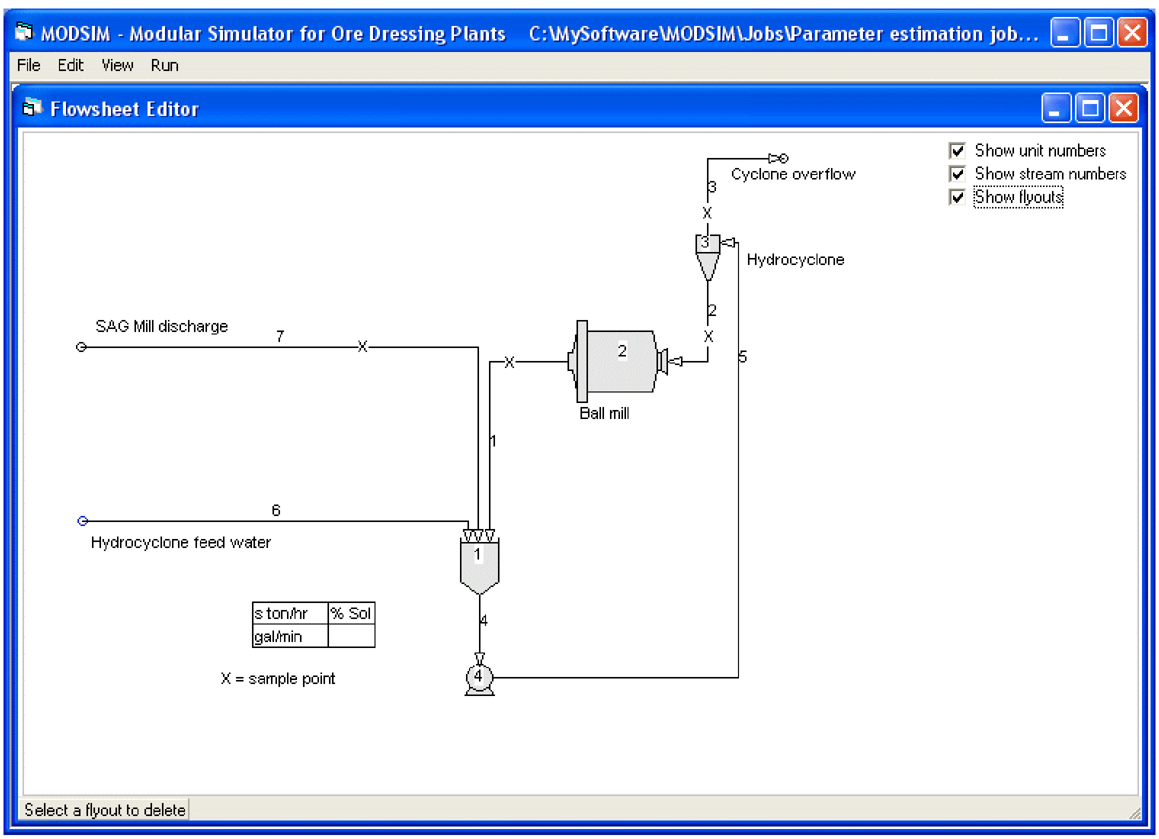

Figure 1 Example of a flowsheet that matches the experimental equipment.

to match the feed stream that was used to generate the data. This should include mass flowrate of solid and water, the particle size distribution, and the appropriate properties of the particles.

An example of an experimental flowsheet is shown in Figure 1. In this example the parameters for the ball mill and hydrocyclone models will be estimated using experimentally determined size distributions in the ball mill feed, ball mill discharge, the hydrocyclone feed and the hydrocyclone overflow. The size distribution in the SAG mill discharge which constituted the feed to this section of the plant was also measured but it cannot be used as fitted data because this size distribution is not influenced in any way by the operation of the ball mill or the hydrocyclone.

The measured size distributions of the product and internal streams must be specified in Modsim in the normal way so that the simulated size distributions can be compared to the experimental data. The data for the ball mill feed in this example is shown in Figure 2.

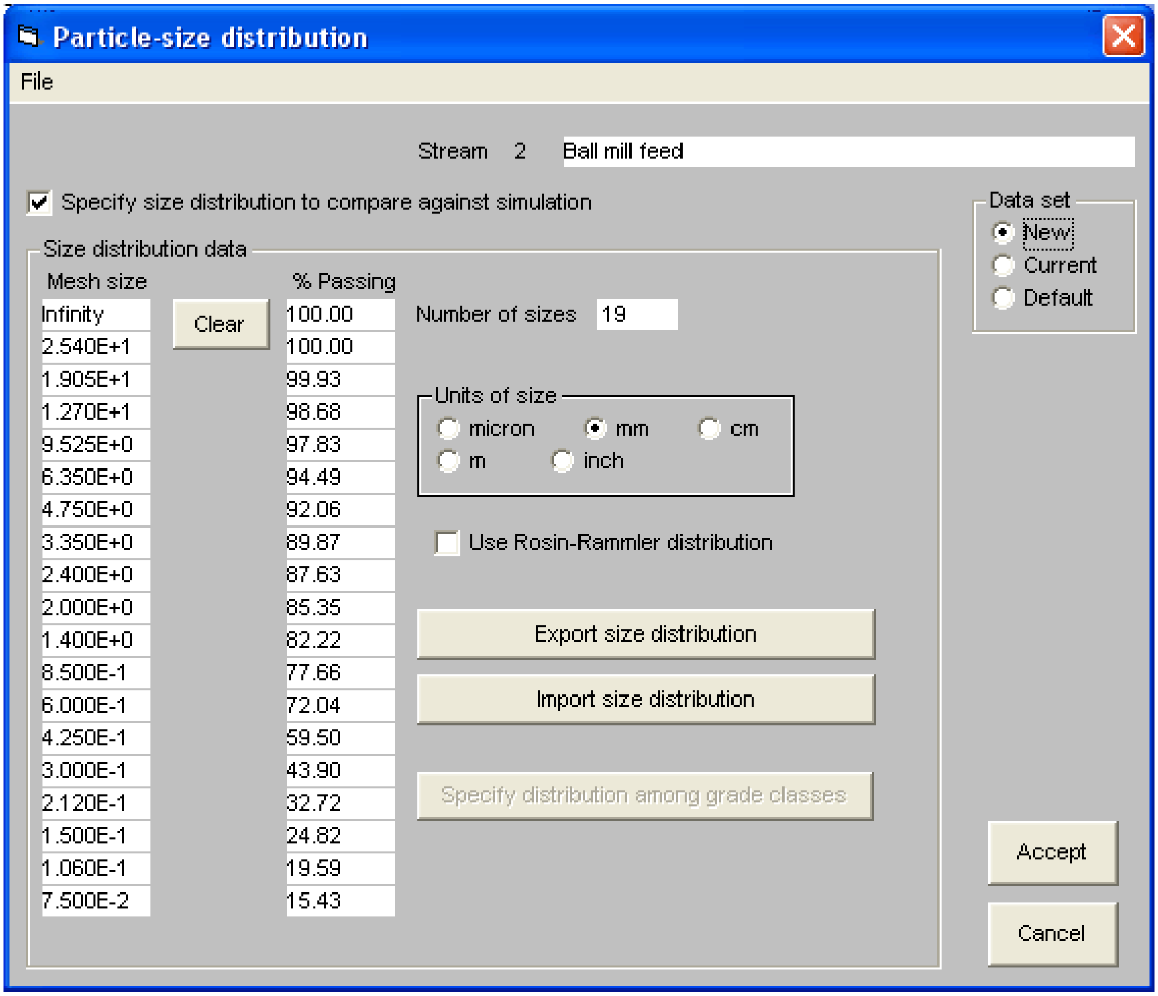

Figure 2 Measured size distribution in the ball mill feed stream.

This MODSIM job must be run and the adjustable unit model parameters should be adjusted to establish a reasonable match with the experimentally measured size distributions. Parameters that are meant to reflect actual physical settings in the equipment should be set appropriately to match the actual experimental conditions. This preliminary simulation serves as the starting point for the parameter

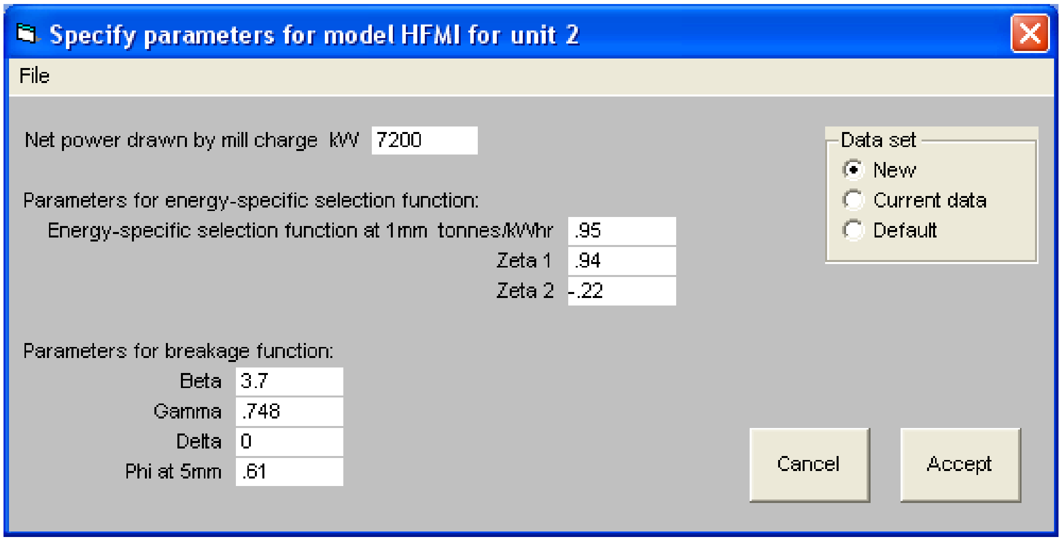

Figure 3 Modsim data input form to specify the parameters for the experimental mill

estimation iterative calculation that is undertaken during stage 2. Because the iterative calculation is difficulty and enormously computationally intensive it can take a relatively long time. Time spent on adjusting the preliminary values of the model parameters is usually amply repaid in time saved during the later iterative calculations.

The parameter specification form for the mill in this example is shown in Figure 3. The parameters that were chosen for estimation were the three parameters S_1^E, ζ_1 and ζ_2 in the energy-specific model for the

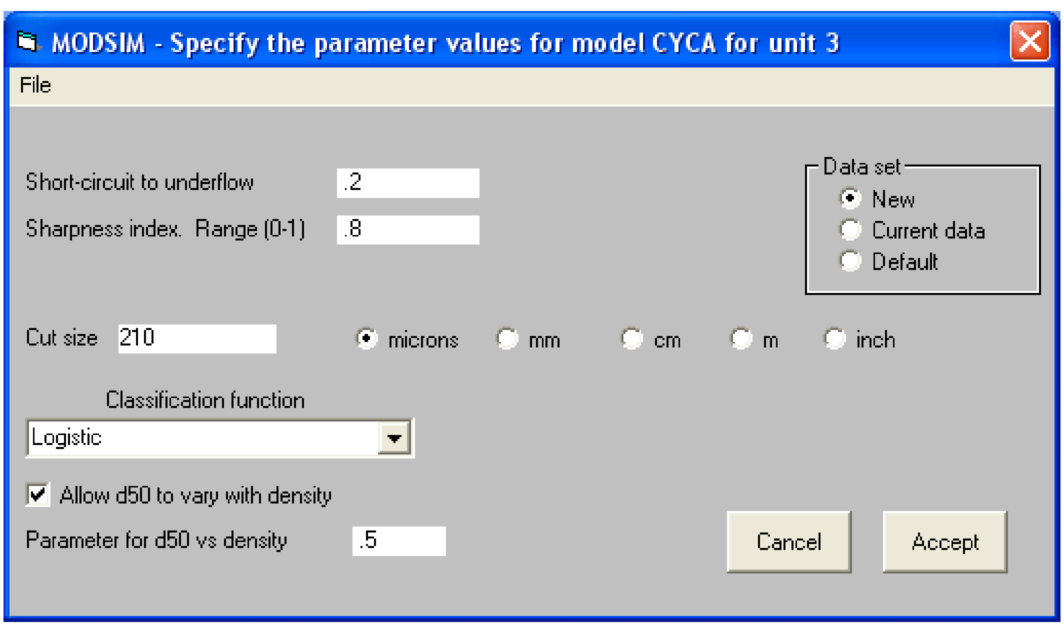

Figure 4 Modsim data input form for the specification of parameters for the hydrocyclone.

specific rate of breakage and the parameter Φ_5 in the model for the breakage function. The power input to the mill was measured during the experimental test and it is therefore not a parameter that is available for estimation.

The parameter specification form for the hydrocyclone model is shown in Figure 4.

The simulated size distribution in the mill discharge using the preliminary parameters is shown in Figure 5.

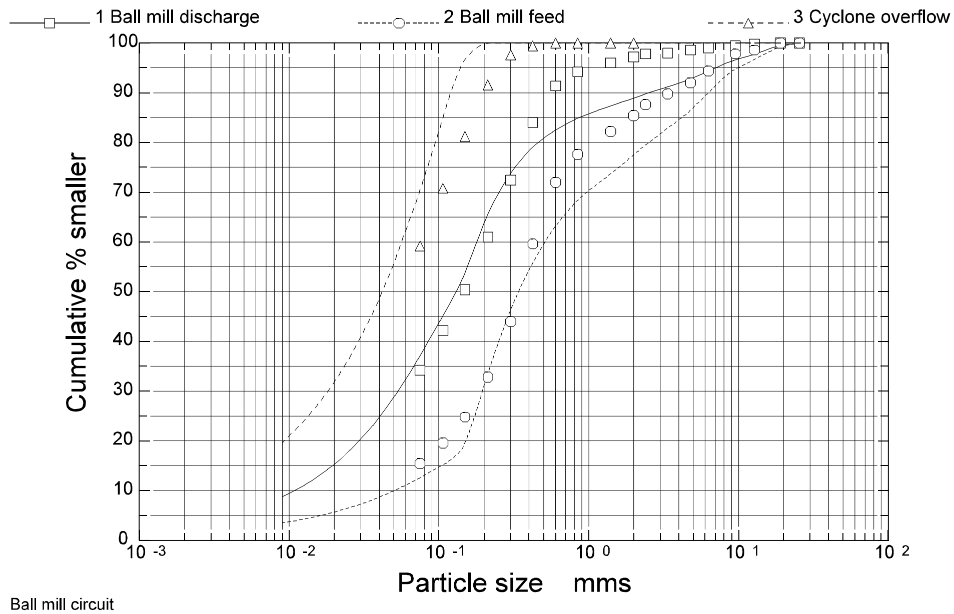

Figure 5 Size distribution from the Modsim simulation using the initial values of the parameters.

The Modsim job has now been set up to start the parameter estimation phase. The Modsim job can now be saved so that it can be opened in the parameter estimation module.

STAGE 2

Start Modsim-Curvefit and open the job that was saved from MODSIM in stage 1. The control window for this program is shown in Figure 6

The first step in the parameter estimation phase is to select the particle size distributions that are to be

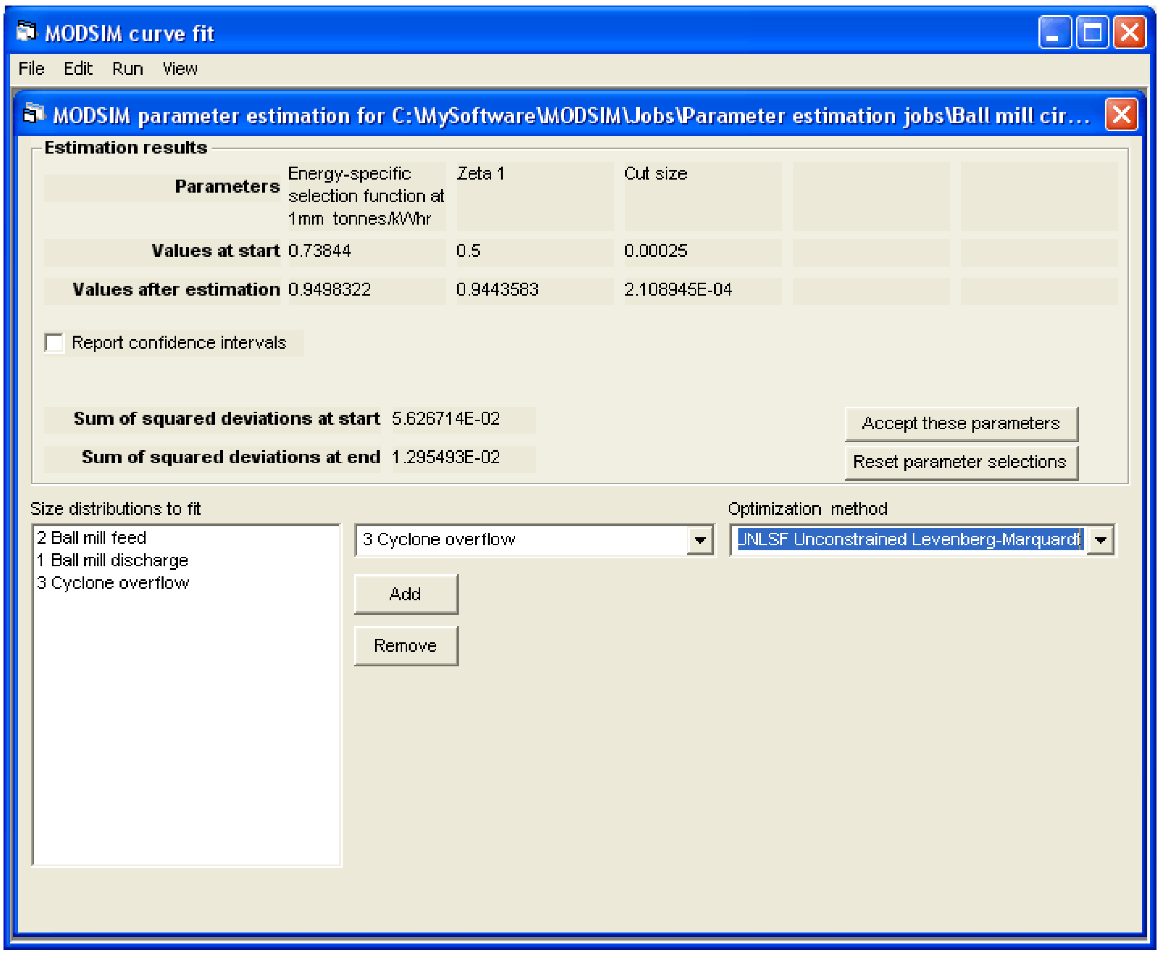

Figure 6 Control window for the parameter estimation program.

matched by the model. Select these from the drop-down list in the Curve-fit window as shown in Figure 6.

The next step is to select the parameters that are to be searched. This is done by selecting `Edit parameters` on the Edit menu. This will bring up the Modsim model parameter screen. Select the unit model from the top of the list and double click the model. The parameter values that are displayed will agree with the values that were used to run the saved job from Modsim in stage 1 because they were imported from the saved Modsim job.

To select a parameter for estimation, click on the data input field for that parameter while depressing the control key. This will bring up a parameter selection window as shown in Figure 7. The starting value for the estimation search can be specified. The existing value is automatically shown and this is normally used. If a constrained search method is to be used, upper and lower values that will constrain the search can also be selected. This is sometimes useful when the search program attempts to search in an infeasible region for the parameter. For example, most parameters cannot take negative values and if these are sent to Modsim by the search engine, Modsim will not converge to a useful solution and the output from Modsim will not be useful for the estimation search. Such weird parameter values could also cause Modsim to fail catastrophically. The search variables are also numbered and a number for this parameter must also be selected. This procedure is repeated for each parameter that will be included in the search. Up to 5 parameters may be chosen for searching. Since multivariable searching is difficult and time consuming, it is usual to search on smaller numbers of parameters until the search has progessed close to ultimate convergence.

Before starting the search, the optimization method that is to be used must be selected. Three options are

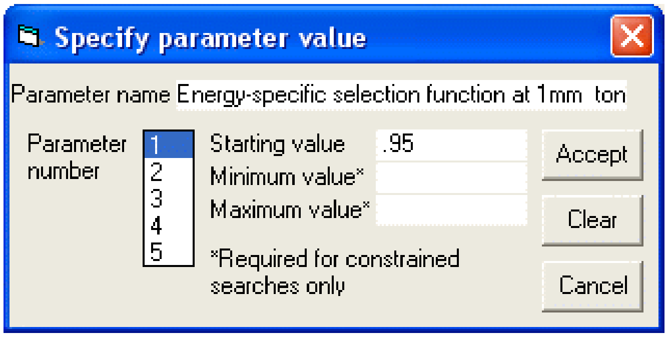

Figure 7 Select the parameter for estimation and specify a starting value.

provided: unconstrained and constrained Levenberg-Marquardt and unconstrained conjugate gradient methods are offered. The unconstrained Levenberg-Marquardt method is commonly regarded as the modt efficient search method for least squares minimization problems but all three methods have been found to give good performance.

The estimator is activated from the Run menu.

The least-squares best estimates are displayed in the result window as shown in Figure6. The sum of squared deviation at the start of the search and at the end point is also shown. This is the best indicator of the success or otherwise of the estimation search.

The quality of the fit between the simulated and experimental size distributions can be seen by viewing the size distributions from the `View` menu.

If the new parameters are acceptable, these should be accepted from the main control window. This will put these values into the Modsim job.

Normally several rounds of parameter estimation are required before the final set of values is found. The parameter selections should be reset from the main control window before a new search is initiated.

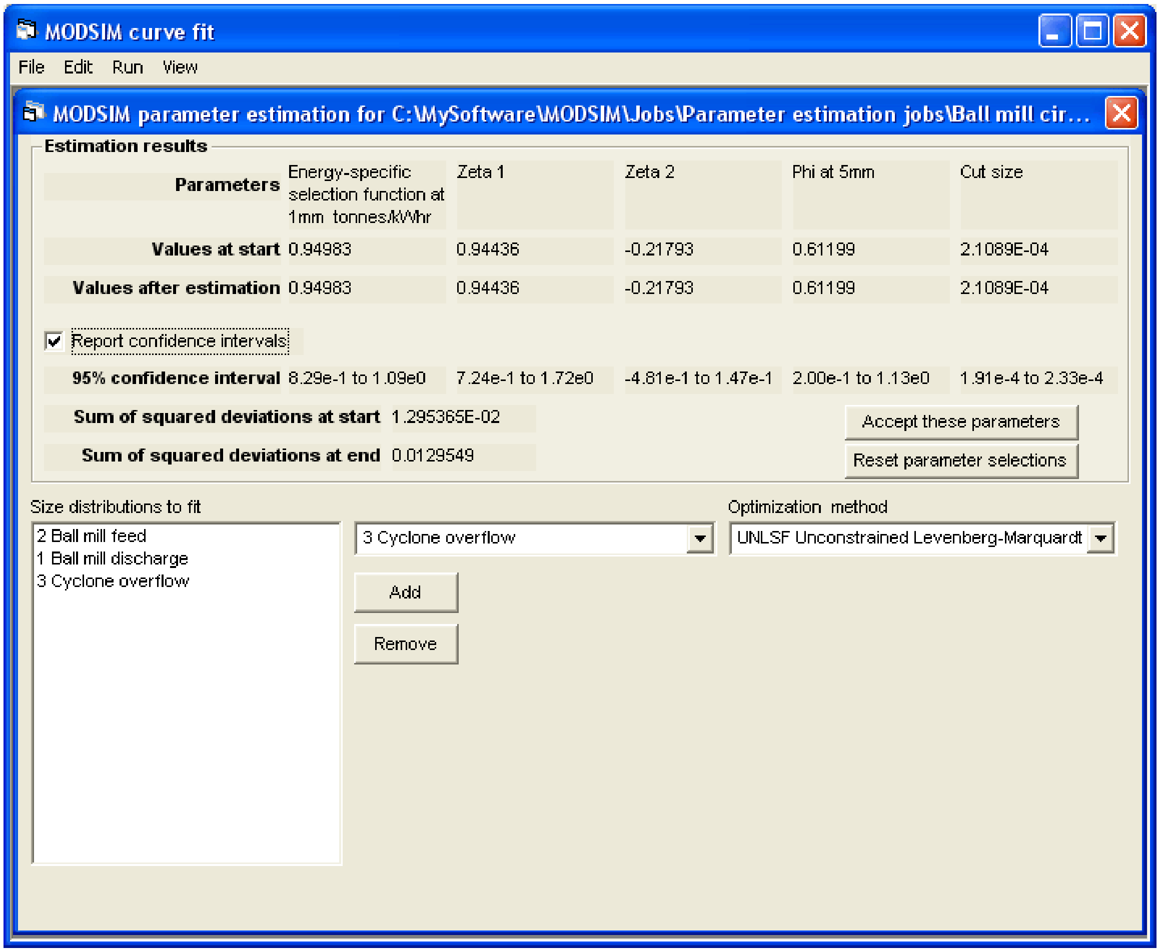

Figure 8 Final results from the parameter estimation showing the 95% confidence intervals for the parameter estimates.

The 95% confidence interval for each parameter estimate can also be obtained by checking the box in the main control form. The calculation of the confidence limits is time consuming so that this is normally done only once the least squares best values of all the parameters have been found. The result of a final run with the confidence intervals is shown in Figure 8.

When the parameter estimation phase has been completed, the job should be saved in the Curve-fit program. The job can be opened in Modsim where it will run with the new parameters. The quality of the data fit can be checked by viewing the size distributions in Modsim. The results of the example are shown in Figure 9.

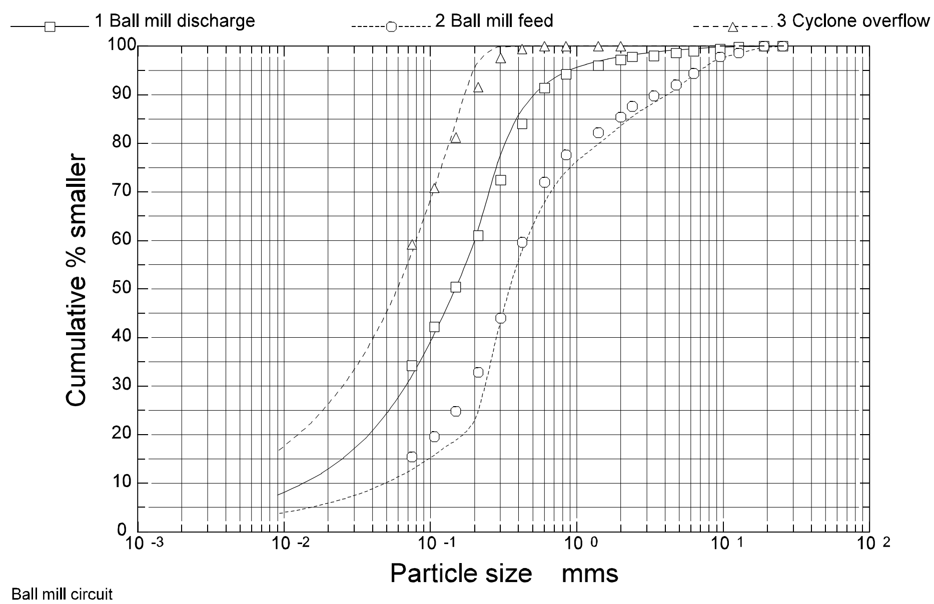

Figure 9 Size distributions calculated in Modsim using the least squares best values of the model parameters that were found by the curve fit estimator.